

Calculus and Analytical Geometry

Exercise 1.4

Q.1

Determine the left hand limit and right hand limit and then find limits of the following functions at $x = c$.

(ii) $f(x) = 2x^2 + x - 5$, $c = 1$

sol:

L.H.L =

$$\begin{aligned}\lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 - 4 \\ &= -2\end{aligned}$$

R.H.L =

$$\begin{aligned}\lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 - 4 \\ &= -2\end{aligned}$$

Hence , $L.H.L = R.H.L$

$$(ii) f(x) = \frac{x^2 - 9}{x - 3}, \quad c = -3$$

Sol:

L.H.L =

$$\begin{aligned}\lim_{x \rightarrow -3} f(x) &= \lim_{x \rightarrow -3} \frac{x^2 - 9}{x - 3} \\&= \lim_{x \rightarrow -3} \frac{(x+3) \cancel{(x-3)}}{\cancel{x-3}} \\&= \lim_{x \rightarrow -3} x + 3 \\&= -3 + 3 \\&= 0\end{aligned}$$

R.H.L =

$$\begin{aligned}\lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} \frac{x^2 - 9}{x - 3} \\&= \lim_{x \rightarrow -3^-} \frac{(x+3) \cancel{(x-3)}}{\cancel{x-3}} \\&= \lim_{x \rightarrow -3^-} x + 3 \\&= -3 + 3 \\&= 0\end{aligned}$$

Hence, L.H.L = R.H.L

(iii) $f(x) = |x-5|$, $c = 5$

Sol:

L-H-L =

$$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} |x-5|$$

$$= \lim_{x \rightarrow 5} -(x-5)$$

$$= -(5-5)$$

$$= 0$$

R-H-L =

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} |x-5|$$

$$= \lim_{x \rightarrow 5^-} +(x-5)$$

$$= 5-5$$

$$= 0$$

Hence, $L-H-L = R-H-L$

Q-2

Discuss the continuity of $f(x)$ at $x = c$:

(i) $f(x) = 2x+5$ if $x \leq 2$, $c = 2$
 $4x+1$ if $x > 2$

Sol:

(i)

$$f(x) = 2x + 5$$

$$f(2) = 2(2) + 5$$

$$= 4 + 5$$

$$= 9$$

(Defined)

(ii)

L·H·L =

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (2x + 5)$$

$$= 2(2) + 5$$

$$= 9$$

R·H·L =

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (4x + 1)$$

$$= 4(2) + 1$$

$$= 8 + 1$$

$$= 9$$

Hence, L·H·L = R·H·S

So, Limit Exists.

(iii)

$$f(2) = \lim_{x \rightarrow 2} f(x) = 9$$

So, function is continuous at $x=2$

$$(ii) f(x) = \begin{cases} 3x - 1 & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2x & \text{if } x > 1 \end{cases}, c = 1$$

sol:

$$(i) f(x) = 4$$

$$f(1) = 4$$

(Defined)

(ii)

L-H-L =

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (3x - 1) \\ &= 3(1) - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

R-H-L =

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2x) \\ &= 2(1) \\ &= 2 \end{aligned}$$

Hence, L-H-L = R-H-L

So, Limit Exists.

(iii)

$$f(1) \neq \lim_{x \rightarrow 1} f(x)$$

So, function is discontinuous at $x=1$

Q.3

$$\text{If } f(x) = \begin{cases} 3x & \text{if } x \leq -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \geq 2 \end{cases}$$

Discuss continuity at $x=2$ and $x=-2$

Sol:

(i) At $x = 2$

$$f(2) = 3$$

(Defined)

(ii)

L.H.L =

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (x^2 - 1) \\ &= 2^2 - 1 \\ &= 3 \end{aligned}$$

R.H.L =

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} 3 \\ &= 3 \end{aligned}$$

Hence, L.H.L = R.H.L

So, Limit Exists.

(iii)

$$f(2) = \lim_{x \rightarrow 2} f(x) = 3$$

So, function is continuous at $x=2$

(i)

$$\text{At } x = -2$$

$$f(-2) = 3(-2) = -6 \quad (\text{Defined})$$

(ii)

$$L.H.L =$$

$$\begin{aligned} \lim_{x \rightarrow -2} f(x) &= \lim_{x \rightarrow -2} (3x) \\ &= 3(-2) \\ &= -6 \end{aligned}$$

$$R.H.L =$$

$$\begin{aligned} \lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} (x^2 - 1) \\ &= (-2)^2 - 1 \\ &= 3 \end{aligned}$$

Hence, $L.H.L \neq R.H.S$
So, function is discontinuous at $x = -2$

Q.4

$$\text{If } f(x) = \begin{cases} x+2, & x \leq -1 \\ c+2, & x > -1 \end{cases}$$

Find "c" so that $\lim_{x \rightarrow -1} f(x)$ exists

Sol:

L-H-L =

$$\begin{aligned}\lim_{x \rightarrow -1} f(x) &= \lim_{x \rightarrow -1} (x+2) \\ &= -1+2 \\ &= 1\end{aligned}$$

R-H-L =

$$\begin{aligned}\lim_{x \rightarrow -1^-} f(x) &= \lim_{x \rightarrow -1^-} (c+2) \\ &= c+2\end{aligned}$$

Since Limit Exists:

$$\text{L-H-L} = \text{R-H-L}$$

$$1 = c+2$$

$$c = 1-2$$

$$c = -1$$

Q-5

Find the values m and n , so that given functions f is continuous at $x=3$.

$$(i) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x+9 & \text{if } x > 3 \end{cases}$$

Sol:

L-H-L =

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (mx)$$

$$= 3m$$

R-H-L =

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (-2x + 9)$$

$$= -2(3) + 9$$

$$= -6 + 9$$

$$= 3$$

Since $f(x)$ is continuous at $x=3$

$$L-H-L = R-H-L = f(3)$$

$$3m = 3, \quad 3 = n$$

$$\boxed{m = 1}$$

$$, \quad \boxed{n = 3}$$

Ans

$$(ii) f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases}$$

Sol:

$$f(3) = (3)^2 = 9$$

L.H.L =

$$\begin{aligned}\lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} (mx) \\ &= 3m\end{aligned}$$

R.H.L =

$$\begin{aligned}\lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (x^2) \\ &= 3^2 \\ &= 9\end{aligned}$$

Since $f(x)$ is continuous at $x=3$

$$L.H.L = R.H.L$$

$$3m = 9$$

$$m = \frac{9}{3}$$

$$\boxed{m = 3}$$

Ans

Q.6

$$\text{If } f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} & , x \neq 2 \\ k & , x = 2 \end{cases}$$

Find value of k so that f is continuous at $x=2$.

Sol:

$$f(2) = k$$

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} \times \frac{\sqrt{2x+5} + \sqrt{x+7}}{\sqrt{2x+5} + \sqrt{x+7}}$$

$$= \lim_{x \rightarrow 2} \frac{(\sqrt{2x+5})^2 - (\sqrt{x+7})^2}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{2x+5 - x-7}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{x-2}}{(\cancel{x-2})(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{1}{\sqrt{2x+5} + \sqrt{x+7}}$$

$$= \frac{1}{\sqrt{2(2)+5} + \sqrt{2+7}}$$

$$= \frac{1}{\sqrt{9} + \sqrt{9}}$$

$$= \frac{1}{3+3}$$

$$= \frac{1}{6}$$

$$= \frac{1}{6}$$

Since Limit is continuous at $x=2$

$$f(2) = \lim_{x \rightarrow 2} f(x)$$

$$\boxed{k = \frac{1}{6}}$$

Ans